Salient Is Not Enough: Keeping Every Time Anchor for Temporal Grounding under a Visual-Token Budget

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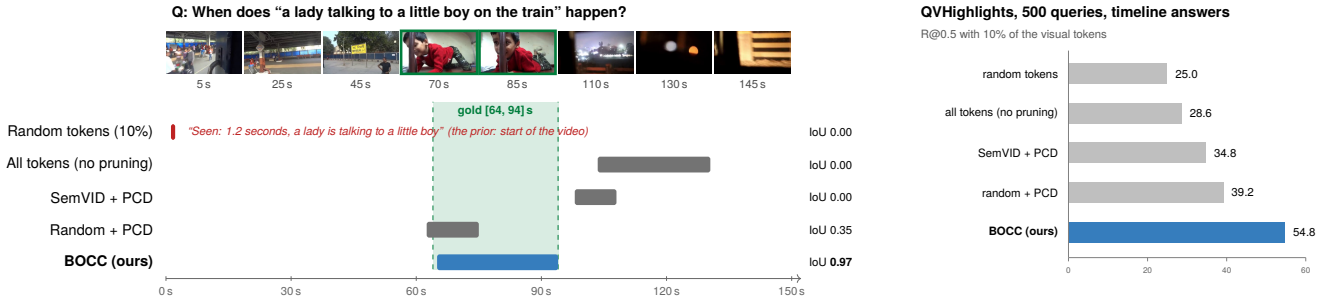


Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers "1.2 seconds"; QA-oriented pruning and the uncompressed model pick a later, wrong scene. BOCC keeps every frame anchored in time and contrasts the answer against the model's no-video prediction, and recovers the interval (IoU 0.97). Right: R@0.5 over 500 queries when every method answers with a timeline of observations, the format of grounded QA. In this format the prior writes many of the times and BOCC is far above the uncompressed model. With a one-line answer the uncompressed model is much stronger, and Coverage alone matches it with a quarter of the tokens (Sec. 6.2).

Abstract

Video LLMs prune most of their visual tokens with little loss in multiple-choice accuracy, but temporal grounding suffers. A timestamp survives only with a visual token of its frame, so QA pruners that empty frames fall below random selection, and in reasoning-style answers a video-free prior fills the gaps. We propose BOCC, a training-free method that selects frames by the question (BOLT), keeps the most question-attended tokens in every frame (Coverage), and contrasts each written time with the model's no-video prediction (PCD). With 31% of the tokens, Coverage alone matches the uncompressed model on QVHighlights and Charades-STA. On grounded QA (S-EMBER) with 10% of the tokens, BOCC improves mIoU over the uncompressed model by up to +11.2. Gains are small on short videos.

1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer gives a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance, and they require knowing when things happened across the whole window.

Video LLMs pay for every frame. Qwen3-VL [38] spends about 330 visual tokens per frame, so 64 frames cost about 21,000 tokens. Pruning has therefore become the default, and many works report that 10–25% of the visual tokens preserve multiple-choice accuracy [14, 18, 27, 35, 41, 44]. We ask what pruning does to *temporal grounding*. In short, **pruning for QA keeps what is salient, but grounding needs what is everywhere.** A QA answer can be read off a few informative frames. An interval needs a timeline with an anchor in every part of the window.

Observation 1: QA pruners lose the anchors. Video LLMs such as Qwen3-VL read time from timestamp tokens interleaved with the visual tokens. A frame that loses all its visual tokens loses its timestamp too, or keeps it with nothing to show. QA pruners drop frames that look redundant, unremarkable or unrelated to the question, which are exactly the frames a timeline needs. All four recent pruners we evaluate fall below random token selection at the same budget on S-EMBER, QVHighlights and Charades-STA (Sec. 6.3).

Random selection leaves a few tokens on every frame, and one published pruner gains 5.3 mIoU just by ranking tokens within each frame instead of globally (Sec. 3.2).

Observation 2: a prior fills the silence. Grounded QA invites reasoning-style answers that list observations and their times before the interval. Asked this way without video, the model still gives confident intervals, and 90% of them start in the first 5% of the window. Pruned memories share the habit: 30–56% of their intervals start there, against 16% of the gold intervals, and removing the evidence frames raises this share from 37% to 51% (Sec. 3.3). The prior is not tied to any pruner. It is what the decoder writes when the memory is silent.

Observation 3: the format decides which failure dominates. On the same QVHighlights queries and memory, asking for a timeline of observations instead of a single interval costs about 30 mIoU (Sec. 6.2). With one-line answers, question-driven token selection helps and contrastive decoding does nothing. With timeline answers, the prior writes many of the times, so contrasting it away helps unless the contrast breaks the format. We therefore treat the answer format as an experimental variable.

We propose BOCC (BOLT + Coverage + Contrastive decoding), a training-free method with one component per failure and an optional BOLT front end that chooses which frames to encode (Fig. 3).

- **Coverage** (Sec. 4.1) keeps the anchors. It prunes the KV cache layer by layer, gives each frame an equal share of the budget, and keeps within each frame the tokens the question attends to most. The quota guarantees every time anchor, and the question decides what each anchor shows. In the settings we tested, a global top-k over the same attention does as well, so the question score drives the gain and the quota is a safeguard against scores collapsing onto a few frames.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2) removes the prior. It chooses each time value by contrasting the model's prediction with its prediction from a video-free memory, as in contextual calibration and context-aware decoding [42, 52], with a strength that fades where the model is confident.

Neither operator is new on its own. Per-frame budgets appear in VidCom^2, KTV and AdaTP [1, 12, 35], question-attention KV eviction in SnapKV and PruneVid [27, 32], and contrast against a content-free or degraded input in contrastive decoding, classifier-free guidance and VCD [19, 30, 31, 40, 42]. We contribute an account of *why* grounding needs each, when each matters, and how to combine them:

1. **A diagnosis of grounding under compression** (Sec. 3): frame-emptying pruners and, shown causally by evidence removal, a video-free prior in reasoning-style answers.
2. **Coverage** (Sec. 4.1). With the one-line prompt on the full Charades-STA and QVHighlights sets, for Qwen3-VL-8B and the grounding-tuned TimeLens-8B, it matches the uncompressed model with 24% of the visual tokens and beats random selection by $+0.5$ to $+1.6$ mIoU (Sec. 6.2). On NExT-GQA at 10% of the tokens it keeps the uncompressed accuracy, while random selection loses about four points. Against a 1 fps input of similar budget, most of the gain comes from the BOLT front end.
3. **PCD and an answer-format study** (Secs. 4.2 and 6.2). PCD helps reasoning-style S-EMBER answers on all eight memories we test and on both models. With 10% of the tokens, BOCC improves on the uncompressed model by $+11.2$ and $+10.7$ mIoU under the same prompt, and by $+5.5$ under the official S-EMBER prompt, where the smaller accuracy gain ($+2.9$) comes mostly from BOLT. PCD does nothing for one-line answers and lowers NExT-GQA grounding where it disturbs the format (TimeLens-8B timeline answers, minutes:seconds under the official prompt).
4. **Reproducible evaluation** (Secs. 5 and 6.2). We reproduce published TimeLens-Bench numbers within 0.3 mIoU, run no-video and shuffled-frame or shuffled-timestamp sanity checks, and report negative results, including the limits of Coverage on short, densely sampled videos.

2. Related Work

Frame selection. BOLT [34] scores frames with a contrastive image–text model and samples them by inverse transform sampling. AKS [43], MDP3 [5], FOCUS [2] and Q-Frame [6] trade relevance against coverage or diversity. All are evaluated on multiple-choice QA. We adapt BOLT to grounding (Sec. 4.3) and find that it adds little once the question chooses the tokens.

Visual token pruning. Query-agnostic methods prune by redundancy or saliency. FastVID [41] keeps density peaks per segment, VidCom^2 [35] budgets frames by uniqueness, FlashVID [18] uses [CLS] attention and tree merging, and DyCoke [44] and FrameFusion [21] merge similar tokens. Query-aware methods use the text. SemVID [14] combines word-embedding relevance with MMR diversity, AdaTP [1] boosts text-like segments, LGTTP [4] reallocates tokens by temporal words, and PruneVid [27] keeps the top 40% of tokens by question attention at one layer. KTV [12], the closest in structure, selects keyframes and then tokens under a relevance-weighted per-frame budget. Only LGTTP evaluates grounding, at a milder budget and offline. BOCC uses an equal per-frame quota to keep every timestamp, and scores tokens by the actual question's attention propagated exactly through all layers. We show that emptying frames

hurts grounding, and that with the actual question's attention the quota is a safeguard and not where the gain comes from (Sec. 6.4).

KV-cache compression and video memory. SnapKV [32], H_2O [51], PyramidKV [15] and AdaKV [20] evict KV entries by attention in long-context LLMs. Our token stage is SnapKV-style eviction for video with a per-frame constraint. For streaming video, HERMES [10] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks when the question arrives, and StreamMem [8] and InfiniPot-V [3] keep bounded memories. All report QA only. We build a memory per question from the frames seen up to the question time. It is causal but not a persistent streaming memory, and we make no claim about its build cost (Sec. 7).

Contrastive decoding and prior correction. Contrastive Decoding [31] subtracts a weaker "amateur" model and introduced the plausibility constraint we use. Classifier-free guidance [40] and context-aware decoding [42] amplify the difference between predictions with and without the context, the form of our Eq. (4). Correcting for a prior measured with a content-free input is older. Contextual calibration [52] rescales label probabilities by the answer to "N/A", and domain-conditional PMI [24] reweighs answers by their task-conditioned prior. In vision-language models, VCD [30], ICD [46] and M3ID [19] contrast against a noised image, a disturbed instruction and no image, PAI [33] amplifies image attention and subtracts text-only logits, and SEASON [7] uses temporally corrupted video negatives. These target hallucination, faithfulness or classification. PCD applies the contrast only to *time values*, with a confidence-adaptive strength, and we study when it helps and when it does not (Secs. 6.2 and 6.5).

Grounding-tuned video LLMs. TimeChat [39], VTimeLLM [25] and LITA [26] train video LLMs to localise moments with timestamp-aware encoders, boundary-aware training or time tokens. TimeLens [50] re-annotates the standard benchmarks and fine-tunes Qwen-VL with reinforcement learning. All use a full visual memory. We use TimeLens-8B and TimeLens-Bench and ask how such a model fares when its memory is pruned.

Training-free temporal grounding. NumPro [48] draws frame numbers on frames, FV-Action [9] scans the video with yes/no questions, REZE [13] recasts grounding as recognition, and [Authors to verify] [11] read intervals from calibrated attention heads, debiased against a blank video. All assume the full set of video tokens. BOCC grounds from a *compressed* memory, and these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

Benchmark biases. Otani et al. [37] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because the annotations carry positional priors.

3. Grounding under a Token Budget: a Diagnosis

This section establishes Observations 1 and 2 on S-EMBER; Observation 3 (answer format) is studied in Sec. 6.2. The reference memory is random 10% pruning unless stated otherwise.

3.1. Setting

A question x is asked at time q about a video $v_{0:q}$. The model encodes $F{=}64$ frames from $[0, q]$ into a KV cache $\mathcal{M}$ of N visual tokens, interleaved with timestamp tokens (Qwen3-VL uses one "`<t seconds>`" per pair of frames). A budget ρ (default $\rho{=}0.1$) keeps ρN tokens per layer, kept timestamps included. The model answers with a short *timeline*:

```
Seen: t_1 seconds, …
Seen: t_k seconds, …
Answer: …
Time: [s, e]
```

and we parse $[s, e]$ from the last line. This format suits grounded QA, since the answer describes what was seen before the interval. Section 6.2 compares it with the one-line format of grounding benchmarks (*"The event happens in s–e seconds"*). We report mIoU and R@$\{0.3, 0.5, 0.7\}$ against the gold interval, and GQ@0.5, the share of answers judged correct with IoU $\geq$ 0.5. An interval is an *early start* if it starts in the first 5% of the window, and a *leak* if it starts early although the gold evidence starts later.

3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom^2, FlashVID and SemVID all score below random selection (Sec. 6.3). Where the kept tokens lie explains why (Tab. 1). HERMES scores by attention plus recency and keeps 27.6% of the last quarter of the window but 2.5–4.7% elsewhere. FastVID keeps one representative per run of similar frames, where a continuing event lives, and keeps tokens from only 25% of the frames in a median question. Random selection never empties a frame. A timestamp helps only if some content of its frame survives, so emptied frames leave holes in the timeline even when the score is sensible. Ranking HERMES's scores within each frame ("stratified", an equal share per frame) improves mIoU by 5.3.

3.3. The prior fills the gaps

The model answers without video. With an empty visual memory, the model still writes a timeline. 89% of its first `Seen:` moments are at 0 s and 90% of its intervals start in the first 5% of the window (an 88% leak). The median interval is 0.45$\times$ the gold length, and 87% of the ends fall

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

Selector	0–10%	10–25%	25–50%	50–75%	75–100%	mIoU	Leak
HERMES	4.7	3.9	2.6	2.5	27.6	18.3	49%
Stratified	8.7	8.7	8.6	8.6	8.7	23.7	**29%**
Random	8.9	8.8	8.6	8.7	8.6	**25.8**	30%

more than half an interval too early. We call this habit the *temporal prior*.

The prior leaks into answers with video. The same signature appears, more weakly, in every memory (Fig. 2). About 30% of intervals start early at every random budget and without pruning, against 16% of gold intervals. The share rises when coverage is lost, to 49% for HERMES and 37–39% for streaming HERMES. The leak follows where the memory is silent, not how many tokens it keeps.

Removing the evidence causes the leak. We drop the uniform frames inside the gold interval ("evidence") or as many frames elsewhere ("control"), keeping every other frame with its true timestamp (Tab. 2). Without the evidence, answers move to the start instead of spreading out, and the leak rises by 14 points for both the unpruned and the random 10% memory. In the HERMES 10% memory, questions whose evidence keeps under 5% of its tokens leak 56% of the time, against 33% when at least 15% survives.

An instruction does not remove it. Adding "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt barely moved the leak (29.5% → 28.0%) or grounding (−0.1 mIoU). In this format the prior acts in the output distribution and must be removed there. We do not yet know whether it appears with single-interval answers. There PCD has no effect (Sec. 6.2; those runs omit the answer reserve, which alone changes S-EMBER grounding by only +0.1), suggesting the prior decides fewer of the written times. **[TODO: no-video answers and leak under the one-line and direct-answer prompts.]**

The end is the weak boundary. Without contrastive decoding, the median end error across memories is 0.40–0.69 of the gold length and the median start error 0.14–0.30. The most frequent failure, 33% of answers, is an interval entirely *before* the gold one (Tab. 26).

4. Method: BOCC

BOCC intervenes where Sec. 3 found the failures. Coverage prunes the memory, prior-contrastive decoding (PCD) corrects the decoder, and a frame-selection front end (BOLT) picks the input frames. No component needs training (Fig. 3).

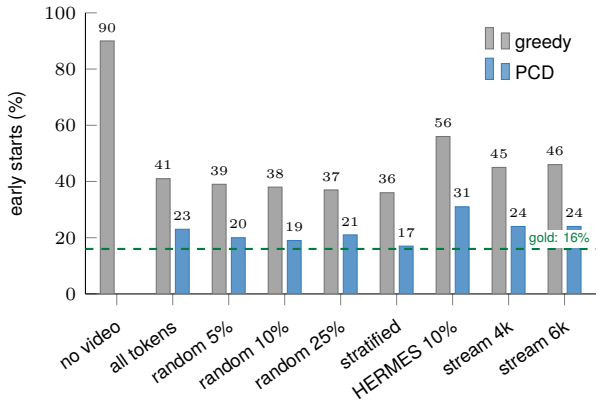


Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2.

		Greedy		PCD	
Memory	Removed	Acc./mIoU/GQ	Leak	Acc./mIoU/GQ	Leak
Random 10%	control	16.6/28.8/8.8	**31%**	15.4/34.6/10.5	**11%**
	evidence	7.8/ 8.6/1.7	46%	9.3/10.3/1.1	23%
All tokens	control	**19.6/29.4/10.1**	37%	**18.1/35.9/12.6**	15%
	evidence	9.1/ 9.4/1.9	51%	9.1/10.5/1.5	22%

4.1. Coverage: equal quotas filled by the question

After the F frames are encoded, we prune the memory once, separately in each of the L layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and B the per-layer budget for visual tokens.

Question relevance. We append the question x to the uncompressed memory and run it through all layers, so the queries at layer ℓ come from the real hidden states. Visual token i is scored at layer ℓ by the attention it receives from the question, averaged over heads h and question positions j:

$$a_\ell(i) = \frac{1}{H\,|x|} \sum_{h=1}^{H} \sum_{j \in x} \mathrm{softmax}_i\Big(\tfrac{1}{\sqrt{d}}\, Q_j^{(\ell,h)}\, K_i^{(\ell,h)\top}\Big). \tag{1}$$

Equal quota. Each frame pair gets $B_f = \lfloor B/F' \rfloor$ tokens,

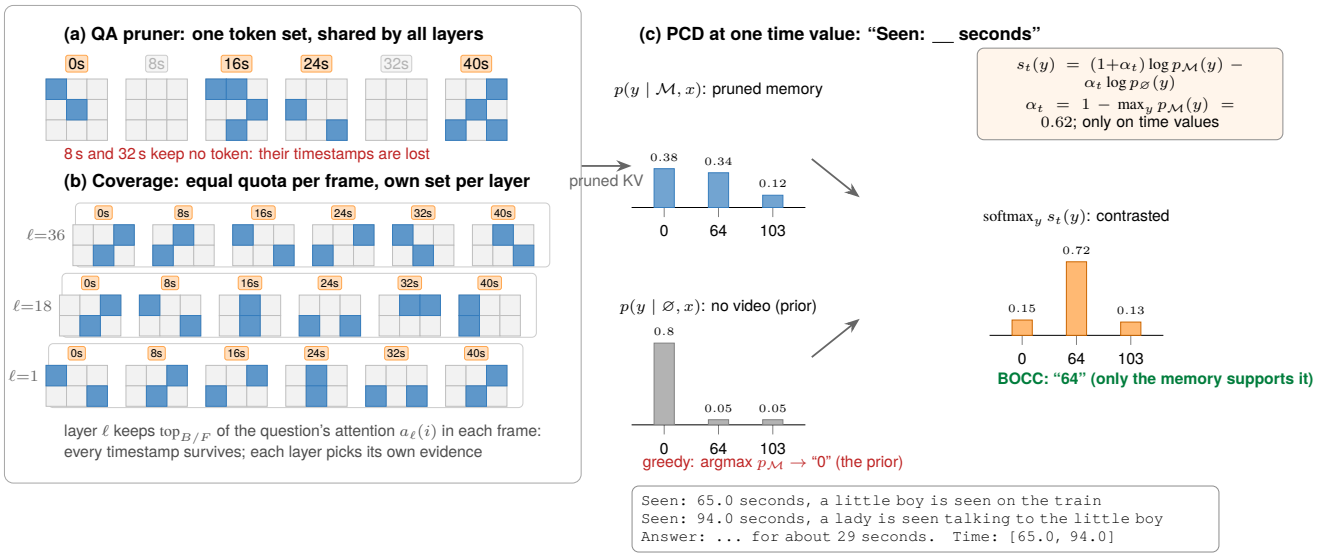


Figure 3. **Overview of BOCC. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 15). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. Output shown is the real BOCC answer.

with any remainder handed out in order, and layer ℓ keeps

$$\mathcal{K}_\ell = \bigcup_{f=1}^{F'} \text{top}_{B_f} \{ a_\ell(i) : i \in \mathcal{T}_f \}. \quad (2)$$

The timestamp tokens of f are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. At any practical budget every frame pair keeps at least one token, so every time anchor survives, while the question decides which part of each frame is shown. Kept tokens keep their original rotary positions and thus their temporal order.

Why equal quotas. The equal quota is a deliberately simple constraint that guarantees the timeline and keeps attention from collapsing onto a few frames. Under attention-mass quotas, sinks and recency bias can give one frame up to 227 of the roughly 2,140 kept tokens, and in a pre-registered test relevance-weighted quotas with a floor did not beat equal quotas (-0.2 mIoU). The quota matters most for generic probe scores as in HERMES [10], which it lifts by 5.3 mIoU. With the actual question, a global top-k without any quota has so far done as well (Sec. 6.4), so there the quota is a guarantee and not the source of the gain.

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. Forcing one shared set by averaging attention over layers changes BOCC by -1.5 mIoU and leaves random selection unaffected ($+0.7$). Per-layer selection thus helps question-driven scores specifically and does not explain why random selection beats input-level pruners.

Frames and budget. With 2–4 tokens per frame pair, the most-attended tokens carry little content and Coverage collapses (Sec. 6.7), so for small budgets we spend the tokens on fewer frames. At a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. We use F=64 when each frame pair keeps at least ∼8 tokens and halve F otherwise. Kept timestamps count against the budget and take a large part of budgets of a few hundred tokens, so some of the collapse may come from this accounting (Sec. 6.6). **[TODO: rule derived from Sec. 6.7; validate on held-out data and with the fixed timestamp rule.]**

Cost. Coverage adds one forward pass of the question over the uncompressed memory per question. Since the full memory is prefilled first, Coverage shrinks the memory the decoder reads but not the cost of building it. PCD decodes a second, prompt-only row in lockstep at every step, which roughly doubles the decoder's per-step work on a much shorter context. **[TODO: measured prefill, time to first token, total latency, peak memory and KV size; the current implementation runs 3–4 scoring passes, reducible to one.]**

4.2. PCD: prior-contrastive decoding of time values

Let $p_\mathcal{M}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_\varnothing(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and gener-

ated prefix but no visual tokens. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory says nothing about y, the evidence term is flat and the video-free prior decides (Sec. 3.3). PCD amplifies the evidence term,

$$\begin{aligned} s_t(y) \;&=\; (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &=\; \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \qquad (4)$$

and picks $y_t = \arg\max_{y\in\mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with β=0.1, as in VCD [30], so tokens the memory makes implausible are never promoted. We adapt this rule to grounding in three ways.

No-video counterfactual. We use an empty negative memory instead of a noised or shuffled one. This isolates exactly the prior of Sec. 3.3, and it is cheap, since its cache holds only the prompt.

Time scope. PCD is active only while the model writes a time value, inside a `Seen:` time and inside `Time: [`s`,` e`]`. All other tokens, including the answer text, are decoded greedily from $p_{\mathcal{M}}$.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\big(1 - \max_y p_{\mathcal{M}}(y)\big)$ with $\alpha_{\max}$=1. When the memory makes the model confident, *e.g.* because the evidence really starts at 0 s, PCD fades out. When the model hesitates between several times, the evidence term decides.

Prompt format and answer reserve. Under the timeline prompt, PCD acts first on the `Seen:` moments and then on the interval, which is thus written from contrasted observations. With PCD, the last 64 tokens of the generation budget are reserved for the `Answer:` and `Time:` lines, and the decoder starts an `Answer:` line once only 64 remain. The reserve alone changes S-EMBER mIoU by only +0.1. A single-interval answer, as in the one-line prompt, leaves PCD little to correct (Sec. 6.2). Under the official prompt we have no clean PCD-only comparison on S-EMBER yet, and on NExT-GQA PCD lowers the scored grounding by writing more times as minutes:seconds (Sec. 6.5).

4.3. BOLT: question-guided frames

When the window is much longer than F frames, choosing the frames can matter as much as choosing the tokens. We sample frames as in BOLT [34]. Each second of video is embedded once with SigLIP 2 [45], giving a causal 1 fps index. The relevance z_j of second $j \leq q$ is its z-scored cosine similarity with the question, and the F frames are taken at evenly spaced quantiles of

$$p(j) \;=\; \tfrac{1}{2}\cdot\tfrac{1}{q} \;+\; \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \qquad (5)$$

so half of the frames remain uniform. We skip BOLT when uniform frames would be less than 1 s apart. With question-driven tokens, BOLT adds only +0.8 to +0.9 mIoU on S-EMBER, QVHighlights and ActivityNet. The runs in the main tables include it.

5. Experimental Setup

Models. Our main model is Qwen3-VL-8B-Instruct [38]. It sees 64 frames at native resolution, sampled uniformly from $[0, q]$ or chosen by BOLT (Sec. 4.3), and decodes greedily or with PCD, at batch size 8 for pruned memories. We generate at most 384 new tokens on S-EMBER (repetition penalty 1.1), 512 on TimeLens-Bench, and 256 (two-query) or 1,024 (joint answer) on NExT-GQA, both with repetition penalty 1.0. All methods share the timeline prompt, except on the full TimeLens-Bench sets (including the QVHighlights and Charades-STA columns of Tab. 3) and in the prompt study of Sec. 6.2, which use the TimeLens-Bench one-line prompt. Sec. 6.2 also adds TimeLens-8B [50], a Qwen3-VL-8B model fine-tuned for grounding, with frames capped at about 1.57 MP on S-EMBER and NExT-GQA. On TimeLens-Bench both models use the protocol input (2 fps, at most 14,336 visual tokens and at least 64 per frame), and BOLT runs use 64 or 128 frames of at most 0.79 MP. Batch size alone changes about half of the greedy outputs and moves a score by up to about two points, so comparisons with the uncompressed model (batch size 1–2) include this noise.

Two clusters. Later runs used a second GPU cluster: most fixed-rule and all 5% NExT-GQA runs, BOLT on the full QVHighlights set, the official S-EMBER prompt, and the TimeLens-8B gold-informed runs. Paired differences that combine both clusters are marked. Runs repeated on both clusters agree within one mIoU point, *e.g.* +0.4 for NExT-GQA without pruning (469 of 510 replies identical) and −0.9 for BOCC on S-EMBER.

Datasets. **S-EMBER** grounded QA [47]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream, so the memory holds only the frames before the question. **QVHighlights** [29] val: 500 queries with a single relevant window, sampled with seed 0 from the 1,020 single-window queries, on videos of about 150 s. **ActivityNet Captions** [28] val_1: 500 queries, one sentence per video, on videos of about 115 s. **Charades-STA** [22] test: 500 queries on videos of about 30 s. For these three datasets the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. **TimeLens-Bench** [50]: the re-annotated Charades-STA, QVHighlights and ActivityNet sets, with their one-line

prompt, on the full sets and on a 513-query QVHighlights subset. **NExT-GQA** [49] test: 1,005 questions on videos of about 40 s, with the answer and the interval asked in two separate queries (multiple choice or open-ended) or, for open-ended answers, in one joint reply. Our QVHighlights and ActivityNet copies are low-resolution re-encodes with 153 and about 84 visual tokens per frame pair, *vs*. 660 on S-EMBER, so we also report budgets in absolute tokens.

Budget. The default budget is 10% of the visual tokens of the 64-frame memory, timestamps included (about 2,140 tokens on S-EMBER, about 520 on QVHighlights). ActivityNet is reported at 30% (400–900 tokens), because at 10% its memory holds only 120–300 tokens (Sec. 6.6). Since timestamps count against the budget, a KV selector keeps fewer visual tokens than an input pruner at the same nominal ρ, about 3.4% on NExT-GQA at 10%. A frame's timestamp is kept when the frame keeps a visual token. An earlier version of this rule could let timestamps fill the budget of very small memories and leave no visual token. The NExT-GQA 10% runs use the fixed rule (Sec. 6.2) and the other runs the earlier one, which can matter only where timestamps take a large share of the budget (Sec. 6.6).

Baselines. We compare against all tokens (no pruning), random tokens, and four QA pruners: FastVID [41], VidCom^2 [35], FlashVID [18] (vision-side stages) and SemVID [14], re-implemented and checked to keep the same tokens as the authors' code on test inputs. Each baseline is run with and without PCD. A global question top-k without quotas is compared in Sec. 6.4. **[TODO: SnapKV-style observation-window eviction, PruneVid, and FV-Action / attention readout on the same memory.]**

Metrics. We report mIoU and R@$\{0.3, 0.5, 0.7\}$. On S-EMBER we also report answer accuracy and GQ@0.5, judged by a local Qwen3.8-27B-FP8 model with the official S-EMBER judge prompt. Verdicts for the same run differ by up to about one accuracy point across our three judging sessions (*e.g*. 21.7 in A *vs*. 21.9 in B for one BOCC run), so we never pair Acc. or GQ@0.5 across sessions. Each table uses one session, named in its caption: A for Tab. 4, B for Tabs. 9 and 23, and C for Tab. 24. The early-phase tables listed in Sec. 6 use their own earlier sessions and are compared only within themselves. Open-ended NExT-GQA answers are judged by the same model with the Video-ChatGPT prompt [36], in separate batches that share a cache. On S-EMBER, times given as minutes:seconds are converted to seconds for every method (Appendix A). The joint NExT-GQA answers are scored with the official S-EMBER parser, which counts such times as missing.

Protocol. We made all design choices (scores, quotas, α, scope) on S-EMBER, then froze the configuration and ran it once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

6. Results

Section 6.1 compares BOCC with all baselines, each dataset under the prompt that gives the methods their best results: the reasoning-style timeline prompt on S-EMBER and the one-line prompt on QVHighlights and Charades-STA. Section 6.2 covers the answer format, a grounding-tuned model and NExT-GQA, and the later subsections analyse pruners, components, budget, gold-informed bounds and remaining errors.

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given, and Δ is a paired difference. Tables from the early S-EMBER phases (Tabs. 16 to 19 and 26 and Tab. 2) use an earlier implementation of PCD, the official parser and batch size 1. On the S-EMBER rows we checked, the two parsers differ by 0.2–0.7 mIoU, except for the oracle selector with PCD (1.5 mIoU, because 17 answers are written in minutes:seconds). All other S-EMBER tables use the minutes:seconds-aware parser. The best value is in **bold** and the second best underlined, per column unless a caption says per row. Δ columns are not ranked.

6.1. Main results

Table 3 supports three conclusions.

(i) QA pruners lose to random selection, and Coverage does not. On S-EMBER all four published pruners score below random tokens (Sec. 6.3), while Coverage on uniform frames scores 31.7 mIoU against 25.6 for random tokens and 28.0 for all tokens. On QVHighlights and Charades-STA the pruners fall below random selection under the timeline prompt (Fig. 4). With the one-line prompt on the full QVHighlights set and BOLT frames at about the same visual budget (bottom block), FastVID and SemVID score 36.1 and 53.1 mIoU, against 55.0 for random tokens and 57.2 for Coverage.

(ii) With the one-line prompt and about 24% of the visual tokens, Coverage keeps the grounding of the uncompressed model on both full sets. It leads random selection by a small margin ($+1.6$ and $+0.5$ mIoU), largest on QVHighlights, whose videos are longer. With BOLT frames on QVHighlights (bottom block), it leads random tokens by $+2.2$ mIoU at 1,192 kept tokens and $+0.6$ at 4,455, where it is 2.5 mIoU below and 0.9 above the uncompressed model, respectively (Sec. 6.2).

(iii) On S-EMBER, with reasoning-style answers, both parts of BOCC matter. BOLT adds 0.8 mIoU to Coverage and PCD another 6.8. BOCC is $+11.2$ mIoU above the uncompressed model in the same format and $+4.2$ above random tokens with PCD (Tab. 4), so the gain does not come from PCD alone. Under the one-line prompt PCD has no effect (Tab. 6), and the one-line BOCC cells are not yet run. Timeline-prompt results on 500-query subsets of the other datasets are in Tabs. 13, 21 and 22. On single-moment

Table 3. **Main results**, mIoU / R@0.5, with Qwen3-VL-8B and the prompt that gives the methods their best results on each dataset. S-EMBER: timeline prompt, 475 questions, 64 frames, 10% of the tokens. QVHighlights and Charades-STA: the one-line TimeLens-Bench prompt on the full sets (1,541 and 3,363 queries) at the protocol input, with 31% of all tokens including timestamps (about 24% of the visual tokens); in these two columns, all tokens is the TimeLens evaluation script. BOLT is skipped on Charades-STA, where uniform frames are already dense (–). The bottom block uses BOLT frames on the full QVHighlights set at other budgets and is not ranked: 64 or 128 frames, keeping 1,192 or 4,455 tokens per query including timestamps. At 64 frames, random tokens and Coverage keep about 970 visual tokens; FastVID and SemVID, which keep every timestamp, were set to the same visual budget and kept 803 and 915 on average. **[TODO:]** cells are not yet run.

Method	S-EMBER timeline, 10%	QVHighlights one-line, 31%	Charades-STA one-line, 31%
All tokens (no pruning)	28.0 / 26.7	59.6 / **64.5**	48.5 / **54.0**
Random tokens	25.6 / 24.6	58.1 / 62.6	48.2 / 52.3
FastVID [41]	17.9 / 15.8	**[TODO:]**	**[TODO:]**
VidCom^2 [35]	20.4 / 18.7	**[TODO:]**	**[TODO:]**
FlashVID [18]	22.7 / 21.5	**[TODO:]**	**[TODO:]**
SemVID [14]	23.3 / 20.6	**[TODO:]**	**[TODO:]**
Coverage (uniform frames)	31.7 / 32.4	**59.7 / 64.5**	**48.6 / 54.0**
BOLT + Coverage	32.5 / 34.1	**[TODO:]**	–
BOLT + Coverage + PCD (BOCC)	**39.3 / 40.8**	**[TODO:]**	**[TODO:]**
BOLT 64 frames + random tokens	–	55.0 / 59.8	–
BOLT 64 frames + FastVID	–	36.1 / 34.2	–
BOLT 64 frames + SemVID	–	53.1 / 57.4	–
BOLT 64 frames + Coverage	–	57.2 / 62.9	–
BOLT 128 frames + random tokens	–	59.9 / 65.3	–
BOLT 128 frames + Coverage	–	60.5 / 66.1	–

Table 4. **Grounded QA on S-EMBER** (10%; all rows from judging session A, Sec. 5). ΔAcc. *vs.* random tokens.

Method	Acc.	GQ@0.5	mIoU	ΔAcc.
All tokens	19.8	8.8	28.0	+2.5
Random tokens	17.3	7.4	25.6	–
FastVID	15.2	4.8	17.9	−2.1
VidCom^2	16.2	5.9	20.4	−1.1
FlashVID	17.3	6.1	22.7	+0.0
SemVID	19.8	7.6	23.3	+2.5
Coverage, uniform frames	**22.5**	11.2	31.7	+5.3
Random + PCD	20.4	12.2	35.0	+3.2
SemVID + PCD°	17.9	9.9	32.3	+0.6
BOCC	21.7	13.3	**39.3**	+4.4
BOCC + coverage gate	22.1	**13.5**	**39.3**	+4.8

° earlier implementation of PCD.

Table 5. **S-EMBER by question type**, mIoU / R@0.5 (10%).

Method	Duration (191)	Counting (184)	Location (100)
All tokens	37.6 / 40.3	30.1 / 25.5	5.9 / 3.0
Random tokens	33.2 / 35.1	29.6 / 26.1	3.5 / 2.0
BOLT + Coverage	43.2 / 46.6	35.3 / 36.4	7.1 / 6.0
Random + PCD	41.0 / 40.8	41.9 / 40.2	**11.0 / 9.0**
SemVID + PCD	38.9 / 36.6	44.6 / **50.0**	8.9 / 8.0
BOCC	**47.6 / 50.8**	**46.9** / 48.4	9.3 / 8.0

grounding that format costs the uncompressed model about 30 mIoU.

Answers. BOCC also improves the answers (Tab. 4). Its accuracy is 21.7, against 17.3 for random tokens and 19.8 for all tokens, and its GQ@0.5 rises to 13.3 (judging session A; session B gives 21.9 and 13.5, Tabs. 9 and 23). Coverage alone answers best (22.5), so PCD trades about one point of accuracy for its grounding gain. Accuracy differences below about two points are within the judge's decoding noise.

[TODO: SemVID + PCD in the same judging session.]

Question types. BOCC gains most on duration and counting questions (Tab. 5). Location questions ("where did I put …") remain essentially unsolved (mIoU $\leq$ 11 for every method), partly because annotators disagree on 62 of the 100 intervals (mean pairwise IoU $<$ 0.6), which caps what any method can score.

6.2. Standard prompt, a grounding-tuned model, and short videos

Apart from the one-line columns of Tab. 3, the results so far use one model and the timeline prompt. Here we test three further settings: the standard one-line prompt of TimeLens-Bench [50], the grounding-tuned TimeLens-8B [50], and the short videos of NExT-GQA [49]. Unless stated otherwise,

Table 6. **Answer format** (QVHighlights, 513 TimeLens-Bench queries, Qwen3-VL-8B, 300 frames, 31% of the tokens), mIoU. One-line: the TimeLens-Bench prompt. Timeline: the prompt of all other tables. The one-line PCD run used no answer reserve, the timeline PCD runs a reserve of 64 tokens (Sec. 4.2); on S-EMBER the reserve alone changes mIoU by +0.1.

	One-line prompt		Timeline prompt	
Selection	greedy	PCD	greedy	PCD
All tokens	60.5	–	30.4	–
Random tokens	58.2	–	**33.4**	43.9
Coverage	**61.7**	61.6	30.7	43.9

paired differences use the same questions. Unpruned runs use batch size 1–2 and pruned runs batch size 8, which alone can move scores by up to about two points (Sec. 5). Some runs come from a second GPU cluster. Repeated runs across the two clusters differ by less than one mIoU point, and we flag every pair that mixes them.

Our pipeline reproduces published numbers. With the TimeLens evaluation script, both models match the published scores on the full Charades-STA, QVHighlights and ActivityNet sets of TimeLens-Bench (*e.g.* 48.5 *vs.* 48.3 mIoU for Qwen3-VL-8B on Charades-STA). Without pruning, our own pipeline agrees with the script on the same queries, within 0.0 to +1.4 mIoU (Tab. 27).

The timeline prompt is costly. Table 6 changes only the prompt. On QVHighlights, asking for a timeline of observations before the interval costs about 30 mIoU (−30.0 without pruning, −31.0 for Coverage). PCD recovers part of this loss (+13.2 on Coverage) but stays −17.8 below the one-line prompt, under which PCD has no effect (−0.1). Those one-line PCD runs used no answer reserve, but on S-EMBER the reserve alone changes grounding by only +0.1. **[TODO: one-line PCD with the 64-token reserve.]** With the timeline prompt, Coverage is not better than random selection here (−2.8 greedy, +0.1 with PCD).

The timeline-prompt results on QVHighlights, ActivityNet and Charades-STA (Fig. 1 and Tabs. 13, 21 and 22) therefore handicap the uncompressed model, and their gains over "all tokens" do not hold against the one-line prompt, which the QVHighlights and Charades-STA columns of Tab. 3 use. PCD corrects the times a reasoning-style answer writes before the interval, so a single-interval answer leaves it little to correct.

Coverage under the standard prompt. With the one-line prompt (Tab. 7, top), Coverage at 24% of the visual tokens beats random selection on both models (+3.5 and +2.0 mIoU), and also SemVID and the TimeLens input at a similar budget. Against the uncompressed model it scores +1.2 mIoU for Qwen3-VL-8B and +0.2 for TimeLens-8B.

BOLT frames on the full QVHighlights set. At about 1.2k tokens per query (Tab. 7, bottom), BOLT with Coverage exceeds a 1 fps TimeLens input capped at 1,365 video tokens by +9.2 mIoU on Qwen3-VL-8B and +6.5 on TimeLens-8B. Most of this gain comes from choosing frames by the question at full resolution. BOLT with random tokens already gains +7.0 and +6.1, and Coverage adds +2.2 and +0.4 over it. The comparison also leaves out the cost of the SigLIP index.

Table 7. **Standard grounding prompt** (QVHighlights, one-line prompt), mIoU / R@0.5. Top: 513 TimeLens-Bench queries, 300 frames at protocol size; Kept: visual tokens in memory (keep ratios count timestamp tokens too: 31% of all tokens is 24% of the visual tokens). Bottom: the full set (1,541 queries) with 64 or 128 frames chosen by BOLT; Kept: tokens per query *including* timestamps. TimeLens input: the TimeLens-Bench input at a lower resolution, capped at the given number of video tokens. Each block is ranked separately.

Method	Kept	Qwen3-VL-8B	TimeLens-8B
All tokens	13,650	60.5 / 64.9	67.1 / 73.1
TimeLens input, 2 fps low-res	4,200	58.0 / 62.0	64.3 / 68.8
Random tokens (31%)	3,271	58.2 / 60.8	65.3 / 70.4
SemVID (31%)	4,232	58.8 / 62.4	–
Coverage (31%)	3,271	**61.7 / 66.7**	**67.3 / 73.9**
All tokens, full set	all	59.6 / 64.5	65.7 / 71.8
TimeLens input, 1 fps low-res	≤1,365	48.0 / 51.5	57.0 / 61.4
BOLT 64 + random (23%)	1,192	55.0 / 59.8	63.1 / 69.7
BOLT 64 + Coverage (23%)	1,192	**57.2 / 62.9**	**63.5 / 70.0**
TimeLens input, 2 fps low-res	≤4,500	57.3 / 61.6	63.3 / 68.4
BOLT 128 + random (43%)	4,455	59.9 / 65.3	**65.3 / 71.1**
BOLT 128 + Coverage (43%)	4,455	**60.5 / 66.1**	**65.3** / 70.8

Against the uncompressed memory, BOLT + Coverage loses grounding at about 1.2k tokens (−2.5 and −2.1 mIoU) and is level at about 4.5k tokens (+0.9 and −0.4). With the protocol input instead of BOLT frames, about 1.2k timestamp tokens use up a budget of 8.2%, no visual token survives, and both selectors collapse (4.6 and 4.2 mIoU). At small budgets, fewer frames are needed (Sec. 6.7).

Full TimeLens-Bench sets. On the complete Charades-STA (3,363 queries) and QVHighlights (1,541) sets (Tab. 8), Coverage at 31% of the tokens is lossless on both models, within −0.2 to +0.1 mIoU. In all four cells it beats random selection (+0.5 to +1.6) and the TimeLens input lowered in resolution to the same budget (+2.1 to +2.6). The gap to random is largest on QVHighlights, whose videos are longer. On Charades-STA, 31% still leaves about 3,350 tokens and random selection is already close. Even random selection beats the low-res input (+0.8 to +2.2), so keeping full resolution and choosing tokens works better than spending the budget on resolution. **[TODO: full ActivityNet set (paused).]**

A grounding-tuned model on grounded QA. S-EMBER

Table 8. **Full TimeLens-Bench sets** (one-line prompt, 31% of the tokens), mIoU / R@0.5. All tokens and low-res: the TimeLens evaluation script, the latter at a lower resolution with a cap of 3,500 / 4,500 *video* tokens on Charades-STA / QVHighlights. Coverage and random keep 3,348 / 4,652 tokens per query *including* timestamps, so the low-res input sees at least as many visual tokens. Paired ΔmIoU below.

	Qwen3-VL-8B		TimeLens-8B	
Method	Charades	QVHighlights	Charades	QVHighlights
All tokens (no pruning)	48.5 / **54.0**	59.6 / **64.5**	**55.4 / 63.1**	**65.7 / 71.8**
TimeLens input, low-res	46.0 / 49.6	57.3 / 61.6	52.8 / 59.6	63.3 / 68.4
Random tokens (31%)	48.2 / 52.3	58.1 / 62.6	54.5 / 61.7	64.2 / 69.6
Coverage (31%)	**48.6 / 54.0**	**59.7 / 64.5**	55.2 / **63.1**	65.4 / 71.0
Coverage *vs.* random	+0.5	+1.6	+0.7	+1.2
Coverage *vs.* low-res	+2.6	+2.4	+2.4	+2.1
Coverage *vs.* all tokens	+0.1	+0.1	−0.2	−0.2

Table 9. **A grounding-tuned model on S-EMBER** (10%, timeline prompt), mIoU / R@0.5 and Acc. / GQ@0.5. All Acc. and GQ@0.5 values are from judging session B (Sec. 5).

	Qwen3-VL-8B		TimeLens-8B	
Method	mIoU / R@0.5	Acc. / GQ	mIoU / R@0.5	Acc. / GQ
All tokens (no pruning)	28.0 / 26.7	19.6 / 8.8	33.1 / 34.1	20.6 / 12.2
Random tokens	25.6 / 24.6	17.1 / 7.2	28.2 / 26.7	19.6 / 9.5
Random + PCD	35.0 / 33.9	20.0 / 12.0	38.4 / 39.4	20.4 / 12.4
BOLT + Coverage	32.5 / 34.1	**22.3** / 11.2	37.2 / 37.9	22.9 / 13.1
BOCC	**39.3 / 40.8**	21.9 / **13.5**	**43.9 / 46.9**	**27.4 / 17.5**
BOCC *vs.* all tokens	+11.2	+2.3 / +4.6	+10.7	+6.7 / +5.3

Table 10. **NExT-GQA, multiple choice** (test, 1,005 questions, 64 frames, two-query protocol: one query for the answer, one for the interval; no PCD). Coverage and random budgets count timestamp tokens, so at 10% they keep about 3.4% of the visual tokens; SemVID, an input pruner, keeps the nominal share (matched comparison: Tab. 11). All tokens: batch size 2, first cluster; 10% and 5% rows: batch size 8, second cluster. † Earlier timestamp rule, which could leave no visual tokens on low-budget questions (about 282 of 2,010 prompts at 5%); reruns with the fixed rule are pending. All other KV-selector rows use the fixed rule or are unaffected (30%).

Model	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Qwen3-VL-8B	All tokens	79.7	38.9	**48.0**	**37.7**
	Random 30%	78.6	38.6	47.5	37.3
	SemVID 30%	78.9	38.5	46.2	35.8
	Coverage 30%	**79.8**	**39.4**	47.9	37.6
	Random 10%	75.7	36.7	46.6	36.4
	Coverage 10%	79.1	38.5	47.4	37.0
	Random 5%†	68.3	27.3	40.5	30.5
	SemVID 5%	74.0	30.0	41.0	30.0
	Coverage 5%†	75.0	32.4	41.6	31.4
TimeLens-8B	All tokens	**76.7**	**41.5**	**52.1**	39.8
	Random 10%	72.8	38.6	50.1	38.9
	Coverage 10%	76.0	40.8	**52.1**	**40.0**
	Random 5%†	66.0	28.7	43.6	34.0
	Coverage 5%†	71.5	33.3	46.0	35.6

answers explain before they ground, so the timeline format is natural there. TimeLens-8B scores above Qwen3-VL-8B in every row of Tab. 9, and both models rank the methods in the same order. With 10% of the tokens, BOCC improves on the uncompressed TimeLens-8B on every metric (+10.7 mIoU, +6.7 accuracy, +5.3 GQ@0.5) and leads random selection by +15.7 mIoU. Each component helps on its own. PCD adds +10.2 mIoU to random tokens and +6.7 to BOLT + Coverage, and BOLT + Coverage adds +9.0 to random tokens.

A second judging session (C), which also scores the gold-informed runs of Sec. 6.8, agrees with session B within 1.1 accuracy points on every TimeLens-8B row, and every difference above keeps its sign. In it, with Qwen3-VL-8B rerun on the second cluster, BOCC scores +5.4 mIoU higher on TimeLens-8B than on Qwen3-VL-8B, against a gap of +2.4 with random tokens, so the grounding-tuned model profits more from the method. Its gold-informed upper bounds remain well above BOCC.

Short videos. NExT-GQA videos last about 40 s, so 64 frames already sample them densely (Tab. 10). At 10%, Coverage is within 0.7 points of the uncompressed model on every metric for both models, while random selection loses about four points of accuracy (−4.0 and −3.9, cross-cluster pairs). Coverage thus beats random selection on Acc@QA (+3.4 and +3.2), but the two differ little on grounding (+0.6 and +1.1 mIoU). At 30%, Coverage is lossless (+0.5 Acc@GQA, −0.2 mIoU) and grounds better than SemVID (+1.8 mIoU). When frames are dense, where tokens are kept matters most for the answer and less for the interval, as on Charades-STA (Sec. 6.7).

The 5% rows still use the earlier timestamp rule. There Coverage leads random selection on Acc@QA by +6.8 and +5.6, but at 10% the fixed rule raised both selectors by about 2.3 mIoU and 2.4 Acc@GQA, so we do not interpret the 5% differences until the reruns finish. SemVID at a nominal 5% keeps about three times as many visual tokens as Coverage. **[TODO: 5% Coverage, random and global top-k with the fixed timestamp rule, both models; TimeLens-8B SemVID and 30%.]** We use the two-query protocol. Asking for the answer and the interval in one reply, as in prior zero-shot reports [16], lowers Acc@GQA by 16.5 for Qwen3-VL-8B and 9.2 for TimeLens-8B, mostly because the model then writes times as minutes:seconds or copies the prompt's example. Absolute numbers are therefore comparable only within a protocol.

Checks and a matched-budget comparison. Table 11 removes or scrambles the video on a subset of NExT-GQA. Without video, accuracy falls by 32.3 points and grounding to 7.2 mIoU, though the options alone still give half of the

Table 11. **NExT-GQA checks** (Qwen3-VL-8B, multiple choice, the 269 questions of the first 50 test videos; 10% rows use the fixed timestamp rule). The three rows below all tokens remove or scramble the video and are not ranked. SemVID 3.4% and 24.9% keep as many visual tokens as Coverage at 10% and 30%. The 10% block is ranked; the 30% rows are references for the paired comparison in the text.

Memory	Acc@QA	Acc@GQA	mIoP	mIoU
All tokens	83.6	40.1	49.0	37.6
No video	51.3	8.2	17.7	7.2
Frames shuffled	83.6	32.3	38.7	30.9
Timestamps shuffled	84.0	27.5	33.3	26.6
Random 10%	80.3	37.2	46.4	34.8
SemVID 10% (nominal)	78.8	35.3	44.7	32.4
SemVID 3.4% (matched)	71.4	31.2	41.4	28.9
Coverage 10%	**84.4**	**40.5**	**49.0**	**37.5**
SemVID 30% (nominal)	81.8	39.0	47.7	36.4
SemVID 24.9% (matched)	81.0	37.5	45.6	34.3

answers. Shuffling the frames leaves accuracy unchanged and lowers mIoU by 6.7. Shuffling only the timestamps also leaves accuracy unchanged but lowers Acc@GQA by 12.6 and mIoU by 10.9, so the model reads the interval from the timestamps but does not need them for the answer. Keeping 100% of the tokens through either selector reproduces all 538 unpruned replies exactly, and batch size 1 instead of 8 moves no metric by more than 0.4.

At the same number of visual tokens, Coverage leads SemVID on every metric, by +8.6 mIoU at 3.4% and +3.1 at 24.9% (Coverage 30% from the full run, restricted to these questions). The fixed timestamp rule matters only for low-budget questions. On the 80 questions whose 10% budget is about 281 tokens, it raises Coverage from 2.4 to 32.4 mIoU; under the earlier rule 85% of the intervals started at 0 s.

Open-ended answers. Without the options (Tab. 12), answering becomes far harder (33.5 *vs*. 79.7 Acc@QA), while grounding is unchanged because the grounding query is the same. Our judge accepts the ground-truth answer for all 1,005 questions and agrees with letter correctness on 99.2% of the multiple-choice answers. Without pruning, Qwen3-VL-8B reaches 17.4 Acc@GQA zero-shot, against 21.4 for TOGA [23], which is trained for open-ended grounded QA and judged with GPT-3.5. Coverage at 10% keeps the answers in both protocols (−0.3 Acc@QA against all tokens in the two-query protocol, a cross-cluster pair, and +0.2 with the official S-EMBER prompt). In the two-query protocol random selection is close as well, and only SemVID is clearly worse (Coverage leads it by +4.2 Acc@QA and +6.2 mIoU). PCD there slightly lowers grounding (−1.5 mIoU on all tokens, and −1.2 on Coverage in a run with the earlier timestamp rule) and barely changes Acc@GQA,

Table 12. **NExT-GQA, open-ended** (Qwen3-VL-8B, 1,005 questions, options removed). Answers are judged correct when they match the ground truth in meaning (LLM judge, Video-ChatGPT prompt [36]); Acc@GQA also requires IoP ≥ 0.5. Two-query: the answer prompt plus the TimeLens grounding query. Joint: one answer with the interval, with the official S-EMBER prompt; joint intervals are scored with the official S-EMBER parser, which counts times written as minutes:seconds as missing. Rows come from separate judge batches of the same judge with a shared cache (a rerun of all tokens reproduces Acc@QA within 0.3). PCD uses the 64-token answer reserve. All 10% rows use the fixed timestamp rule (Tab. 10). All-tokens rows: batch size 2, first cluster; two-query 10% rows: batch size 8, second cluster; official-prompt rows: first cluster. § Rerun on the second cluster, whose own no-PCD rerun scores 33.2 / 17.1 / 48.0 / 37.5. Blocks are ranked separately.

Protocol	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Two-query	All tokens	**33.5**	17.4	**48.0**	**37.7**
	All tokens + PCD§	33.1	16.6	45.9	36.1
	Random 10%	32.2	16.7	46.7	36.4
	SemVID 10%	29.1	13.7	42.5	31.0
	Coverage 10%	33.2	**17.7**	47.5	37.1
Joint, S-EMBER prompt	All tokens	38.7	13.2	35.1	30.4
	All tokens + PCD	**40.0**	13.4	33.4	28.9
	Random 10%	39.5	12.8	34.4	29.7
	Coverage 10%	38.9	**13.5**	**35.9**	**30.6**
	Coverage 10% + PCD	39.1	13.2	33.6	28.2

as with the one-line prompt on QVHighlights.

Official S-EMBER prompt. This prompt asks for the answer and the interval in one reply, without a list of observations. Coverage and random selection at 10% are both level with all tokens (+0.3 and −0.6 mIoU). PCD raises accuracy slightly on all tokens (+1.3) but lowers mIoU (−1.5 on all tokens, −2.5 on Coverage). Most of this loss is a format effect. PCD more than doubles the answers that write times as minutes:seconds (22 to 58 on all tokens), which the official parser scores as missing. In every run under this prompt, 68–76% of the intervals still start at 0 s.

TimeLens-8B, open-ended. Here we have grounding results only. The answers are not yet judged, and the 10% runs use the earlier timestamp rule. In a joint answer with the timeline prompt, only 22–30% of its intervals start at 0 s. PCD lowers grounding by −4.4 mIoU on all tokens, −4.5 on Coverage and −1.5 on random tokens. TimeLens-8B often ends with an answer and no `Time:` line, and PCD adds run-on or truncated replies, so replies without a parsable interval (scored as IoU 0) rise from 37 to 112. In the two-query protocol, Coverage and random tokens at 10% are level (+0.1 mIoU), and both lose about 2 mIoU against all tokens. **[TODO: judge the TimeLens-8B open-ended answers; fixed-rule reruns of the PCD 10% rows.]**

On NExT-GQA, then, the method contributes through question-driven token selection. PCD does not raise

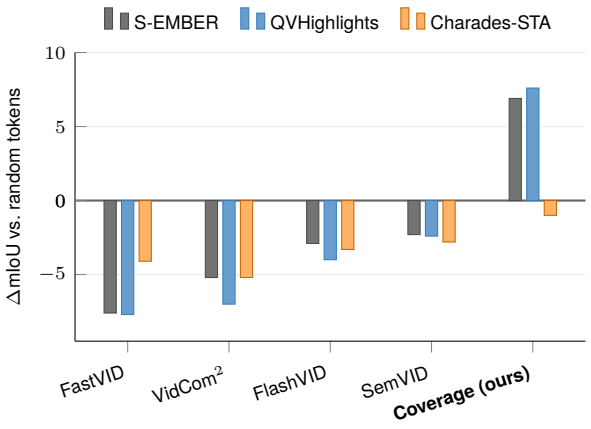


Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets. Coverage gains on the two long-video datasets and is 1.0 mIoU below random selection on 30 s Charades videos.

Acc@GQA in either protocol of Tab. 12, and where it disturbs the answer format, as with TimeLens-8B or with minutes:seconds under the official prompt, it lowers the scored grounding.

6.3. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on three of the four datasets, by -2.3 to -7.6 mIoU on S-EMBER, -2.4 to -7.7 on QVHighlights and -2.8 to -5.2 on Charades-STA. The exception is ActivityNet at 10%, where the input-level pruners are above random (Sec. 6.7). The gaps exceed seed noise (random selection gives 26.0 ± 0.6 mIoU over three seeds on S-EMBER), and their ordering follows coverage (Tab. 1). Pruners that empty frames most aggressively lose the most, and those that spread tokens the least. Accuracy barely moves. FastVID is 2.1 points below random selection in accuracy, about our decoding noise, but 7.6 lower in mIoU. These pruners preserve answers, as designed, but discard the timing. PCD helps every pruner ($+7$ to $+12$ mIoU) but does not close the gap to BOCC.

6.4. Analysis of Coverage

Contribution. Table 13 adds the components one at a time. On the memory side, Coverage gives the largest gain over random tokens ($+6.1$ mIoU on S-EMBER), and BOLT adds only $+0.8$ to $+0.9$ on top. Once tokens are chosen by the question, the encoded frames matter less. **[TODO: Coverage + PCD without BOLT on all datasets.]**

Design choices. Table 14 varies one choice at a time. Given a quota, an equal one works as well as any weighted one. Five relevance-weighted variants (SigLIP similarity, attention mass or both) are all within ± 0.5 mIoU of it, including our pre-registered primary test, the uniform-frame SigLIP variant. Quotas that follow raw attention mass are pulled by attention sinks and recency bias, so that one frame received up to 227 of about 2,140 kept tokens. The score itself should come from the actual question. HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without one. Deterministic top-k beats sampling from a coverage-floored density (31.7 *vs.* 30.0), and a contrastive token score that subtracts a generic sentence's attention does not help.

Is the quota needed? For the generic HERMES score the equal quota is worth $+5.3$ mIoU, because that score concentrates on the end of the window and empties earlier frames (Tab. 1). With the actual question's attention, a global top-k of the same per-layer attention with no quota matched the equal quota in both settings we tested (-0.1 mIoU on S-EMBER under the official prompt, Tab. 20, and -0.0 on the 584 QVHighlights queries run so far with BOLT frames and the one-line prompt, where it is $+1.8$ above random tokens). There, the gain over random selection comes from scoring tokens by the question, and the equal quota guarantees that every anchor survives, which matters for scores that collapse onto a few frames. **[TODO: global top-k without quota under the timeline prompt on S-EMBER and on the complete QVHighlights set (queued); share of frames**

Table 13. **Component ladder** (mIoU / R@0.5), timeline prompt, 10% of the tokens (ActivityNet: 30%); QVHighlights and ActivityNet: 500-query subsets. PCD: prior-contrastive decoding.

Configuration	S-EMBER	QVHighlights	ActivityNet
Uniform, random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6
BOLT, random tokens	29.4 / 29.5	28.7 / 28.0	–
Coverage (uniform frames)	31.7 / 32.4	33.7 / 33.0	24.6 / 21.6
BOLT + Coverage	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2
BOLT + Coverage + PCD	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**

Table 14. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: U = uniform. Δ *vs.* the equal quota with question top-k on the same frames.

Token selection	Frames	mIoU	Δ
Equal quota, question top-k	BOLT	32.5	–
Quota ∝ SigLIP relevance (floor 50%)	BOLT	**33.0**	+0.5
Quota ∝ SigLIP (floor 25%)	BOLT	32.6	+0.1
Quota ∝ SigLIP + attention	BOLT	32.9	+0.4
Quota ∝ attention mass	BOLT	32.1	−0.4
One token set shared by all layers	BOLT	31.9	−0.6
Equal quota, question top-k	U	31.7	–
Quota ∝ SigLIP relevance	U	31.4	−0.2
Equal quota, sampling instead of top-k	U	30.0	−1.7

Table 15. **Per-layer *vs*. shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

Method	Per layer	Shared	ΔmIoU
Random tokens	25.6	26.3	+0.7
Random + PCD	35.0	35.8	+0.7
BOLT + Coverage	32.5	31.9	−0.6
BOCC	**39.3**	**37.8**	−1.5

left empty by it.]

Per-layer selection. Sharing one token set across all layers (Tab. 15) does not change random selection, so per-layer diversity is not why random selection beats input-level pruners. For BOCC it costs −1.5 mIoU, because layers attend to different tokens and question-driven selection benefits from letting each layer choose.

6.5. Analysis of PCD

PCD on other memories. With reasoning-style answers, PCD can be added to any memory. On all eight memories of Tab. 16 it improves mIoU (+4.6 to +8.0) and GQ@0.5 and roughly halves the early starts, gaining most where the memory is smallest. With random selection at 10% it already beats the uncompressed model (32.6 *vs*. 28.0). On FastVID, the pruner that loses the most coverage, it adds +7.6 mIoU and cuts the leak from 38% → 12%.

Choice of counterfactual. Table 17 changes only what the memory is contrasted against. The no-video input works best because it isolates the prior, so contrasting against it removes the prior (early starts 38% → 19%). Noised frames, as in VCD, still show part of the video *and* still produce the prior, so the two runs disagree mainly about content (+0.5 mIoU). Permuted timestamps also leave the prior in place. A one-sided rule that only penalises tokens the video argues against is worse than PCD (−5.6 and −4.8), because without video the model is about 99% sure of "0" and even a well-supported "0" looks like evidence against. Boosting visual attention, or stating the bias in the instruction, does not remove the prior either.

Time scope and adaptive strength. Contrasting every token also changes words unrelated to time. The answer text changed in 85% of questions, and over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 18, top). A constant strength also hurts questions whose evidence really starts at 0 s, lowering their accuracy on the uncompressed memory from 21.8 to 16.7. The adaptive strength fades when the model is confident and raises it to 23.1 instead (Tab. 18, bottom). It also gives the best *discrimination* (early starts on early-evidence questions minus those on the others), 56.4 against 49.6 for greedy decoding on the all-tokens memory.

Other decision rules. We replayed other decision rules offline on both distributions, logged at the first time value of every answer (Tab. 19). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones, so PCD is close to what decoding alone can do. In the remaining early-start errors the memory itself is unsure, and fixing them needs more evidence in the memory.

Prompt format. Under the official direct-answer prompt of S-EMBER (Tab. 20), BOCC still grounds better than the uncompressed model (+5.5 mIoU with the minutes:seconds-aware parser), but its accuracy gain is smaller (+2.9) and comes mostly from BOLT. Adding PCD to BOLT + Coverage raises mIoU by +2.8 and changes accuracy by −0.4, so under this prompt PCD still moves the interval toward the evidence but no longer changes the answers. PCD also makes the model write more times as minutes:seconds, so the official parser cannot read 73 BOCC answers (13–23 for the greedy rows) and hides part of the gain: with it, BOCC is +2.1 above the uncompressed model and PCD changes mIoU by −0.8. The same format effect lowers PCD's scored grounding on NExT-GQA under this prompt (Sec. 6.2). Coverage is level with global question top-k without a quota and with SnapKV-style eviction. Random selection answers 2.5 points below the uncompressed model, every other pruned memory at most 0.6 below; these comparisons carry the batch-size difference noted in the caption. BOCC with the official prompt is 4.7 mIoU below BOCC with the timeline prompt. **[TODO: PCD with and without BOLT under this prompt; the batch-matched timeline-prompt checks and the held-out split (queued).]**

An *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`) states extents and so improves the greedy baseline by +7.4 mIoU. PCD still adds +2.1 to +3.3, reaching 35.0 mIoU at 768 tokens. The two fixes overlap. The format corrects extents, and PCD corrects the leak. BOCC uses the simpler timeline in all main results. **[TODO: event timeline with BOCC.]**

When does PCD help? PCD seems to help in proportion to how much the video-free prior decides the written time: most where the memory says least (Tab. 16), and not at all with a one-line answer (Sec. 6.2; those runs had no answer reserve). It can also change the *form* of the answer, as with minutes:seconds above. On NExT-GQA it lowers TimeLens-8B's timeline-format grounding (−4.4 mIoU) through more replies without a parsable interval. So PCD helps when the prior chooses the times and the answer keeps its format. This is a hypothesis, which the per-token log-probabilities we already store can test directly. **[TODO: PCD gain against the divergence between $p_{\mathcal{M}}$ and $p_{\varnothing}$ at each written time value, across all conditions; one prediction stated before the run.]**

Table 16. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier implementation of PCD). Plain PCD: on all tokens with constant α=0.5; PCD: confidence-adaptive strength on time values only (ours). Δ PCD *vs.* greedy. Best and second best per row.

Memory	Greedy	Plain PCD	PCD	ΔmIoU	ΔGQ@0.5	Early starts
Random 5%	15.8 / 23.6 / 5.9	15.4 / 29.2 / 9.1	**16.4 / 31.5 / 9.9**	+8.0	+4.0	39% → 20%
Random 10%	15.2 / 25.8 / 7.2	**16.2** / 32.1 / **10.1**	**16.2 / 32.6 / 10.1**	+6.8	+2.9	38% → 19%
Random 25%	17.7 / 28.2 / 8.2	**18.9** / 33.0 / 11.8	**18.9 / 34.9 / 13.3**	+6.7	+5.1	37% → 21%
All tokens	18.5 / 28.0 / 8.6	19.8 / 31.5 / 11.2	**21.5 / 32.7 / 13.3**	+4.6	+4.6	41% → 23%
HERMES 10%	13.1 / 18.3 / 4.0	**15.4** / 22.6 / **7.2**	13.3 / **24.1** / 5.9	+5.7	+1.9	56% → 31%
Stratified 10%	14.1 / 23.7 / 6.1	14.5 / 29.9 / 8.8	**15.8 / 31.0 / 10.5**	+7.3	+4.4	36% → 17%
HERMES streaming 4k	13.5 / 21.6 / 4.0	**17.1** / 27.4 / **8.2**	15.4 / **28.6** / **8.2**	+6.9	+4.2	45% → 24%
HERMES streaming 6k	14.3 / 22.6 / 4.8	14.7 / 28.1 / 7.6	**15.4 / 29.8 / 8.4**	+7.2	+3.6	46% → 24%
Mean of 8	15.3 / 24.0 / 6.1	16.5 / 29.2 / 9.3	**16.6 / 30.7 / 10.0**	+6.7	+3.9	

Table 17. **What to contrast against** (S-EMBER, random 10%).

Counterfactual	mIoU	GQ@0.5	Early
None (greedy)	25.8	7.2	38%
No video (ours)	**32.1** (+6.2)	**10.1** (+2.9)	**19%**
Noised frames (VCD [30])	26.3 (+0.5)	7.8 (+0.6)	30%
Evidence-against rule, α=1 / 2	26.5 / 27.3	–	–
Attention gain (PAI-style)	+1.3	+0.8	42%
Instruction "do not assume early"	25.7	6.9	37%

Table 18. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs.* time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

Memory	All tokens	Time values only
Random 10%	**16.2** / 32.1 / 10.1	16.0 / 32.1 / **10.3**
Random 25%	**18.9** / 33.0 / 11.8	17.7 / **34.0** / 11.8
All tokens	19.8 / 31.5 / 11.2	**20.8 / 32.9 / 12.4**
HERMES 10%	**15.4** / 22.6 / 7.2	14.7 / **24.3** / **7.4**
Streaming 4k	**17.1** / 27.4 / 8.2	16.2 / **29.2** / **9.3**
Early-evidence questions (78), all tokens memory		
Greedy	21.8 / 49.2 / 16.7	
PCD, constant α	16.7 / 55.3 / 12.8	
PCD, adaptive α	**23.1 / 56.9 / 19.2**	

Table 19. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

Rule	Early evidence ↑	Other (leak) ↓
Greedy	**69%**	16%
PCD, α=0.5	46%	**6%**
PCD, adaptive	51%	7%
Evidence-against, α=4	51%	9%
Skewed or tempered prior	50–55%	8–11%
Best threshold (uses gold)	46%	**6%**

Table 20. **Official S-EMBER prompt** (direct answer with `Answer:` and `Time:` lines; Qwen3-VL-8B, 64 frames, 10%, second cluster; judge session C). mIoU / R@0.5 with the minutes:seconds-aware parser; mIoU with the official parser and the number of answers it cannot parse. The uncompressed model ran at batch size 4 (8 ran out of memory), the pruned rows at 8.

Method	mIoU / R@0.5	Acc. / GQ@0.5	Official mIoU	Unparsed
All tokens	29.1 / 28.4	13.7 / 4.8	27.4	20
Random tokens	27.9 / 26.1	11.2 / 3.6	26.4	17
Coverage, uniform frames	30.1 / 29.3	13.7 / 5.5	28.8	**13**
Global question top-k, no quota	30.0 / 29.3	13.9 / 5.1	28.5	17
SnapKV-style eviction	30.9 / 29.5	13.1 / 5.1	29.6	16
BOLT + Coverage	31.9 / 32.2	**17.1** / 5.9	**30.2**	23
BOCC	**34.6 / 34.3**	16.6 / **6.3**	29.5	73

6.6. Budget

Table 21 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection stays flat or even improves as the budget *shrinks*. On S-EMBER, BOCC rises from 36.8 at 20% to 39.3 at 5%, while random selection and SemVID degrade. Pruning by the question thus removes distractors as well as cost. At 5%, BOCC leads SemVID + PCD by +7.4 mIoU.

Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair, the top-attended tokens carry little content, and BOCC collapses (9.7 and 11.6 mIoU) below random selection and SemVID, whose cross-frame merging degrades more gracefully. What matters is the absolute number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies. One caveat applies. These runs count the kept timestamps inside the budget under the earlier timestamp rule, which on NExT-GQA left some low-budget questions (about 280 tokens) with no visual tokens (Sec. 6.2), so part of the collapse may come from this accounting rather than from the equal quota. **[TODO: QVHighlights 5% and 10% at 64 frames with the fixed timestamp rule (queued).]**

Table 21. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

	S-EMBER			QVHighlights			ActivityNet		
Budget	5%	10%	20%	5%	10%	20%	10%	30%	50%
Kept tokens	∼1,070	∼2,140	∼4,280	∼260	∼520	∼1,040	120–300	400–900	600–1,500
Random	23.9	25.6	27.4		26.9		17.8	22.9	23.5
SemVID	20.9	23.3	26.5		24.5		21.2	22.8	22.3
R + C (no PCD)	34.6	32.5	31.0		34.5		11.3	25.4	25.0
SemVID + PCD	31.9	34.8	35.3		34.8		**26.8**	30.2	31.4
Random + PCD	34.5	35.0	36.5		37.8		21.5	31.4	32.2
BOCC	**39.3**	**39.3**	**36.8**	9.7	**48.8**	49.4	11.6	**34.7**	**34.4**

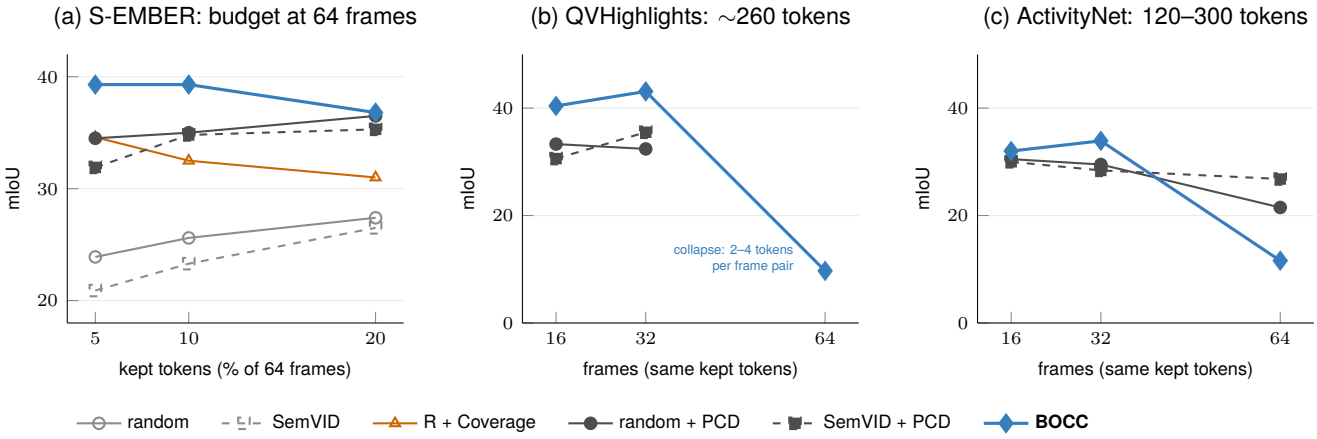


Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; BOCC leads at every budget that leaves at least about eight tokens per frame pair, and collapses below it (QVHighlights 5%, ActivityNet 10%; these runs use the earlier timestamp rule, Sec. 6.6). **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and BOCC collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9).

6.7. Fewer frames or fewer tokens per frame?

The collapse at small budgets comes from how the budget is spent, and the equal quota is not at fault in itself. Each block of Tab. 22 keeps the token count fixed and changes only the split between frames and tokens per frame (the timestamp caveat of Sec. 6.6 applies, since fewer frames also means fewer timestamps in the budget). With 32 frames instead of 64, each frame pair keeps twice as many tokens and BOCC recovers, from 9.7 to 43.1 mIoU on QVHighlights and from 11.6 to 33.9 on ActivityNet, close to its 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little, as temporal coverage becomes the bottleneck. The baselines gain less from fewer frames, and on Charades-STA, whose videos last 30 s, fewer frames help no method. We therefore recommend the rule of Sec. 4.1: keep at least about eight tokens per frame pair, and reduce F when the budget does not allow it. At equal tokens, more frames with fewer tokens each do not help either: 128 frames at 5% on S-EMBER change mIoU by $+1.2$ without PCD and -3.6 with it (74–78 unparseable answers).

6.8. Comparison with gold-informed upper bounds

Table 23 gives the memory information it should not have. On Qwen3-VL-8B, BOCC beats perfect token selection at the same budget without any gold information ($+4.4$ mIoU over the oracle selector with PCD). Knowing which tokens to keep does not stop the prior, and keeping only the gold frames' tokens starves the rest of the timeline. Against the strongest oracle, which sees only the evidence and also uses PCD, BOCC is 2.2 mIoU lower and answers worse (-5.3 accuracy). Even the gold window leaks 45%, so with every frame inside the evidence the model still often starts near 0 s. There PCD lowers the leak to 35% and adds $+5.5$ mIoU, while accuracy changes by -2.3. For Qwen3-VL-8B, more evidence raises accuracy and PCD fixes the time values.

For TimeLens-8B (Tab. 24), the bounds are much higher and BOCC is further from them, though it still exceeds the oracle selector with PCD ($+0.5$ mIoU, $+4.8$ accuracy). The gold window alone is $+13.2$ mIoU above all tokens (both

Table 22. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: F frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

Dataset (kept tokens)	Frames × keep ratio	Random + PCD	SemVID + PCD	BOCC
QVHighlights (∼260)	64 × 5%	–	–	9.7 / 6.6
	32 × 10%	32.4 / 29.2	35.5 / 33.6	**43.1 / 46.6**
	16 × 20%	33.3 / 32.4	30.6 / 28.0	**40.4 / 41.0**
ActivityNet (120–300)	64 × 10%	21.5 / 18.2	**26.8 / 23.6**	11.6 / 7.6
	32 × 20%	29.5 / 26.8	28.4 / 25.8	**33.9 / 32.0**
	16 × 40%	30.5 / 26.8	30.0 / 27.6	**32.0 / 29.8**
Charades-STA (∼10% of 64)	64 × 10%	**26.2 / 24.0**	21.7 / 19.2	23.3 / 21.2
	32 × 20%	**24.6 / 22.0**	20.8 / 16.0	23.6 / 21.0[§]

[§] Coverage on uniform frames; BOLT is skipped on Charades-STA.

Table 23. **Upper bounds that use the gold interval** (S-EMBER, Qwen3-VL-8B, 475 questions; Acc. and GQ@0.5 from judging session B; minutes:seconds-aware parser). Oracle selector: the gold frames' tokens are kept first. Gold window: all 64 frames inside the gold interval (batch size 3). Leak: intervals starting in the first 5% of the window, on the 397 questions whose evidence starts later.

Memory	Gold	Acc.	mIoU	R@0.5	GQ@0.5	Leak
Random 10%	–	17.1	25.6	24.6	7.2	29%
Random 10% + PCD	–	20.0	35.0	33.9	12.0	14%
Oracle selector 10%	✓	18.9	27.9	25.7	9.3	33%
Oracle selector 10% + PCD	✓	18.5	34.9	35.2	9.5	16%
BOCC 10%	–	21.9	39.3	40.8	13.5	**11%**
All tokens	–	19.6	28.0	26.7	8.8	33%
Gold window	✓	**29.5**	36.1	38.1	**15.8**	45%
Gold window + PCD	✓	27.2	**41.5**	**43.2**	14.9	35%

Table 24. **Gold-informed upper bounds for TimeLens-8B** (S-EMBER, 475 questions; all Acc. and GQ@0.5 from judging session C, Sec. 5). Settings as in Tab. 23; all tokens at batch size 1. No oracle selector without PCD was run.

Memory	Gold	Acc.	mIoU	R@0.5	GQ@0.5
Random 10%	–	18.5	28.2	26.7	9.1
Random 10% + PCD	–	19.8	38.4	39.4	11.8
Oracle selector 10% + PCD	✓	21.7	43.3	45.3	14.3
BOCC 10%	–	26.5	43.9	46.9	17.1
All tokens	–	19.8	33.1	34.1	11.6
Gold window	✓	28.2	46.6	46.7	19.2
Gold window + PCD	✓	**36.2**	**54.9**	**57.1**	**25.9**

at batch size 1), and PCD on it adds +8.3 mIoU and +8.0 accuracy. BOCC stays 11.0 mIoU below this bound. For the grounding-tuned model, PCD and evidence thus work together, and the gap shows how much a better choice of frames could still add.

Table 25. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

Idea	Reference	ΔmIoU
VCD-style noised-frame contrast	greedy	+0.5
Permuted-timestamp contrast	greedy	+0.9
Evidence-against decoding rule	PCD	−5.6
Attention gain while decoding	greedy	+1.3
Bias stated in the instruction	greedy	−0.1
Relevance-guided frame zoom	greedy	−1.5
Zoom inside the gold window	greedy	+2.3
Unequal per-frame quotas (5 variants)	equal quota	−0.4 to +0.5
Contrastive token score	question top-k	−1.2 to +0.5
Prior-subtracted retrieval	SigLIP	AUROC ≤ +0.003
128 frames at the same budget	64 frames	+1.2 / −3.6

6.9. Analysis of BOLT

The SigLIP 2 index does find the evidence. Over the seconds before each question, evidence seconds reach an AUROC of 0.663 against the rest, and the top-ranked second lies in the gold interval for 63% of questions (chance 36%) and 41% of location questions (chance 13.5%). BOLT frames put 40% of the memory inside the evidence, against 36% for uniform frames, which helps random tokens (+3.8 mIoU). Once tokens are chosen by the question, BOLT adds only +0.8 to +0.9, because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666), as expected for a contrastively trained retriever, whose score already estimates pointwise mutual information [45]. Re-weighting the token budget of uniform frames by the same relevance adds +1.3. The choice of encoded frames matters more than how the budget is split among fixed frames.

6.10. What did not work

Table 25 collects the ideas that gave little or nothing. They suggest that at a fixed budget, redistributing evidence by zooming, re-weighting frames or unequal quotas is close to its ceiling. Large gains came only from removing the prior

Table 26. **Error types** on S-EMBER, pooled over eight memories, greedy *vs*. PCD (earlier implementation of PCD).

Outcome	Greedy	PCD	Δ
IoU ≥ 0.5, answer correct	6.1%	9.9%	+3.8
IoU ≥ 0.5, answer wrong	16.3%	20.7%	+4.4
Partial overlap (IoU < 0.5)	17.0%	15.7%	−1.3
Inside gold, too short	22.2%	18.5%	−3.7
Entirely *before* gold	33.1%	18.9%	−14.2
Entirely after gold	2.7%	5.7%	+3.0
Covers gold, too long	2.2%	2.5%	+0.3
No answer / unparsed	0.5%	8.1%	+7.6

where it acts (PCD), or from selecting tokens by the actual question in every layer, with or without the equal quota (Sec. 6.4).

6.11. Where answers still fail

Table 26 shows that PCD works where it was aimed. Intervals entirely before the gold one, the signature of the prior, are the largest error type under greedy decoding, and PCD cuts them to less than a fifth of answers. The remaining failures point to four limits. (i) Wrong answers on good intervals. Twice as many answers are well grounded but wrong as grounded and right. Counting answers ignore the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) Recent events. Of the intervals that still end before the gold one, 81% have evidence in the second half of the window, and the model picks an earlier similar event instead. With evidence in the last third, GQ@0.5 is 4.4. (iii) Sparse evidence. With fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more. No decoding rule can recover evidence missing from the input. (iv) Label ambiguity. On the 98 questions whose annotators disagree (mean pairwise IoU < 0.6), every method scores near zero.

7. Limitations

Answer format. The S-EMBER results and most analysis tables use the timeline prompt, which costs the uncompressed model about 30 mIoU on single-moment grounding relative to the one-line prompt (Tab. 6). The gains of BOCC over the uncompressed model on our QVHighlights, ActivityNet and Charades-STA subsets (Tab. 13) hold only for this format. Under the one-line prompt we have run only Coverage and random selection on the full Charades-STA and QVHighlights sets (Tabs. 3 and 8), and nothing yet on the full ActivityNet set. Under the official S-EMBER prompt, BOCC still grounds better (+5.5 mIoU with our parser, +2.1 with the official one, which cannot read 73 of its answers), but its accuracy gain is smaller (+2.9), comes mostly from BOLT, and the uncompressed model ran at batch size 4 *vs*. 8 (Tab. 20). **[TODO: S-EMBER without pruning under the one-line prompt; batch-matched comparison.]**

Where the prior is measured. We measured the video-free prior (Sec. 3.3) only in the timeline format. PCD has no effect on one-line answers (in runs without the answer reserve), which suggests that the prior decides few or none of those times, but we have not measured no-video answers in that format. **[TODO: no-video answers under the one-line and direct-answer prompts.]**

What drives Coverage. The equal quota guarantees every timestamp and helps scores that concentrate on a few frames, such as the HERMES probes (+5.3). With attention from the actual question, a global top-k without the quota matched it in both settings we tested, S-EMBER with the official prompt and a partial QVHighlights run with BOLT frames. Our evidence that grounding needs anchors in every frame thus comes from selectors that empty frames (the QA pruners and generic probe scores) and does not rest on the quota itself. **[TODO: global top-k without quota under the timeline prompt; frame coverage of question top-k.]**

Minimum tokens per frame. Equal quotas fail when a frame pair keeps only 2–4 tokens. Using fewer frames avoids this (Sec. 6.7), but our rule of at least about eight tokens per frame pair comes from those same results and needs held-out validation. The small-budget runs also use the earlier timestamp rule, which left some NExT-GQA questions with no visual tokens, so part of the collapse may be an accounting effect. **[TODO: small-budget runs with the fixed timestamp rule.]**

Short, densely sampled videos. On Charades-STA uniform frames are less than a second apart, and on NExT-GQA 64 frames already cover a 40 s video. There question-driven selection beats random selection only slightly in grounding (+0.5 and +0.7 mIoU on Charades-STA, +0.6 and +1.1 on NExT-GQA at 10%), but at 10% on NExT-GQA it keeps the uncompressed accuracy while random selection loses about four points. The 5% NExT-GQA results use the earlier timestamp rule and are being rerun. On these videos PCD does not raise grounded accuracy in either protocol of Tab. 12 and lowers the scored interval in both. **[TODO: 5% NExT-GQA with the fixed timestamp rule.]**

Where PCD hurts. PCD can change how time values are written as well as which. On NExT-GQA it lowers TimeLens-8B's timeline grounding by −4.4 mIoU because more replies lack a parsable interval, and under the official prompt Qwen3-VL-8B writes more times as minutes:seconds, which the official parser scores as missing. Deployment would need a parser that accepts both forms or a format check on the contrasted reply.

Upper bounds and the front end. BOCC remains well below the gold-window upper bound of TimeLens-8B with PCD (−11.0 mIoU, −9.7 accuracy), and for Qwen3-VL-8B

it is below the gold window in accuracy (-5.3). With BOLT frames, most of the gain over a low-resolution input comes from frame selection and resolution, which random tokens on the same frames also get. We do not count the cost of the SigLIP index.

Not a streaming memory. We rebuild a causal memory for each question from the preceding frames and prefill it in full, so peak memory and time-to-first-token match those of the uncompressed model. BOCC reduces the memory the decoder reads but not the cost of building it, and PCD decodes a second, prompt-only row at every step. **[TODO: cost table; amortisation over several questions per video; write-once memory re-ranked at question time.]**

Scope of the evaluation. Both models come from one family (TimeLens-8B is fine-tuned from Qwen3-VL-8B) and read time from text timestamps. Most diagnostic results use subsets of 475–500 questions, judged by a local model whose sessions differ by up to about one accuracy point, and some paired differences mix two GPU clusters. All hyperparameters were chosen on the S-EMBER questions, and we have not yet evaluated a held-out S-EMBER split. **[TODO: a second model family; held-out S-EMBER split.]**

8. Conclusion

Pruning for QA keeps what is salient, but grounding needs what is everywhere. A video LLM reads time from timestamp anchors that survive only with their frames, so QA pruners, which empty unremarkable frames, fall below random selection on grounding. Keeping in each layer the tokens that the actual question attends to preserves the timeline. With a quarter of the visual tokens this matches the uncompressed model under the standard grounding prompt on the full Charades-STA and QVHighlights sets, and with a tenth it keeps NExT-GQA accuracy for both a general and a grounding-tuned model. When the answer lists what was seen before it states the interval, a video-free prior writes the times wherever the memory is silent. Contrasting each time value with the model's own no-video prediction removes this prior, so that with a tenth of the tokens BOCC grounds better than the uncompressed model answering in the same format, although the contrast can lower the scored grounding when it disturbs the answer format. The answer format alone changes single-moment grounding by about 30 mIoU and should be reported in every evaluation of compressed video memories.

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A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`. The official S-EMBER parser scores such times as missing, so we convert `h:mm:ss` and `m:ss` to seconds for every method. The official parser does not change the method rankings. It gives BOCC in Tab. 13 (timeline prompt) 44.6 / 50.2 on QVHighlights (*vs*. 48.8 / 54.8) and 33.3 / 31.0 on ActivityNet at 30% (*vs*. 34.7 / 32.4). **[TODO: full table with the official parser.]**

B. Checks against published numbers

We reproduce the published unpruned TimeLens-Bench scores with the TimeLens evaluation script and check that our own pipeline matches the script on the same queries (Tab. 27).

Table 27. **Reproduction of TimeLens-Bench** (mIoU, no pruning). Top: TimeLens script on the full sets *vs*. published. Bottom: our pipeline *vs*. the TimeLens script on the same queries (paired Δ).

Model	Dataset (queries)	Ours	Reference	Δ
Qwen3-VL-8B	Charades-STA (3,363)	48.5	48.3 (published)	
Qwen3-VL-8B	QVHighlights (1,541)	59.6	59.4 (published)	
Qwen3-VL-8B	ActivityNet (4,500)	47.1	46.8 (published)	
TimeLens-8B	Charades-STA (3,363)	55.4	55.2 (published)	
TimeLens-8B	QVHighlights (1,541)	65.7	65.5 (published)	
TimeLens-8B	ActivityNet (4,500)	53.2	53.2 (published)	
Qwen3-VL-8B	Charades-STA (400)	48.0	47.5 (script)	+0.5
Qwen3-VL-8B	ActivityNet (400)	47.1	45.7 (script)	+1.4
Qwen3-VL-8B	QVHighlights (513)	60.5	60.5 (script)	0.0
TimeLens-8B	QVHighlights (513)	67.1	66.1 (script)	+1.0

C. Reproduction

BOCC corresponds to the following overrides of our code, which we release with the paper:

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time
(BOLT) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```